5	RNA primer is removed by DNA polymerase I (5'->3' exonuclease activity)	RNA primer is removed by DNA polymerase beta
6	Okazaki fragments are long	Okazaki fragments are short
7	Replication is very rapid	Replication is very slow
8	DNA gyrase is needed to remove the supercoils in the circular bacterial DNA	DNA gyrase is not needed

6.2.9 The polymerase chain reaction – an essential tool for molecular biology

Polymerase chain reaction (PCR) is a method used to make billions of copies of a small region of DNA. This process is also known as PCR amplification. Depending on the application the DNA region being amplified can vary. Some examples include

1. A gene whose function a researcher wants to understand
2. A genetic marker used by forensic scientists to match crime scene DNA with suspects.
3. A genetic marker region in DNA which could be used for diagnosis of a disease like cancer

The goal of PCR amplification is to make enough quantity of the target DNA so that it could be processed for further analysis including sequencing, electrophoresis or plasmid cloning. PCR is routinely used in many areas of biology and medicine, including molecular biology research, medical diagnostics, ecology etc.,

6.2.9.1 The steps involved in PCR amplification

This reaction is essentially a DNA replication carried out in a tube. The key ingredients of a PCR reaction are the enzyme Taq polymerase, primers, template DNA, and deoxy ribo nucleotides (DNA building blocks). The Taq Polymerase is the DNA polymerase enzyme of the heat tolerant bacterium *Thermus aquaticus*. This enzyme can be functional even in temperatures around 70°C. The ingredients are assembled in a tube, along with cofactors needed by the enzyme and are subjected to repeated cycles of heating and cooling that allow DNA to be synthesized. Each cycle of amplification consists of three steps, which are

i. Denaturation (96°C): Heat the reaction strongly to separate or denature the DNA strands. This provides single-stranded template for the next step.

ii. Annealing (55 - 65°C): Cool the reaction so the primers can bind to their complementary sequences on the single-stranded template DNA.

iii. Extension (72°C): Raise the reaction temperatures so Taq polymerase extends the primers, synthesizing new strands of DNA.

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